

PROJECT PHASE - I

RESOURCES



<https://arxiv.org>



<https://mermaid.live>

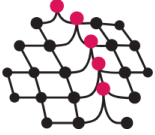


<https://github.com>



<https://www.overleaf.com>

DATA SOURCES

PhysioNet 
<https://physionet.org/>

 **Microsoft Research Open Data**
<https://www.microsoft.com/en-us/research/tools/>

 **Hugging Face Dataset**
<https://huggingface.co/datasets>

Google Research
<https://research.google/resources/datasets/>

 **THE CANCER**
IMAGING ARCHIVE
<https://www.cancerimagingarchive.net/>

 **OpenNEURO**
<https://openneuro.org/>

DOMAINS

1. Theoretical Computer Science and Mathematics

- Computational Complexity
- Computational Geometry
- Computer Science and Game Theory
- Cryptography and Security
- Data Structures and Algorithms
- Discrete Mathematics
- Information Theory
- Numerical Analysis
- Symbolic Computation

2. Computer Systems and Networking

- Databases
 - Distributed, Parallel, and Cluster Computing
 - Hardware Architecture
 - Networking and Internet Architecture
 - Operating Systems
 - Performance
 - Systems and Control
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DOMAINS

3. Artificial Intelligence and Machine Learning

- Artificial Intelligence
- Computation and Language
- Computer Vision and Pattern Recognition
- Information Retrieval
- Machine Learning
- Multiagent Systems
- Neural and Evolutionary Computing
- Robotics

4. Programming Languages and Software Engineering

- Formal Languages and Automata Theory
 - Logic in Computer Science
 - Mathematical Software
 - Programming Languages
 - Software Engineering
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Month 1: Topic and Papers Selection

Week	Goal	Specific Tasks	Deliverable
1	Paper selection	Identify 3 candidate papers (2022+ preferred). Check: open data? open code? feasible in 4 months?	Shortlist with feasibility matrix
2	Finalize paper	Get faculty approval. Download datasets.	Approved paper + dataset
3	Deep technical read	Annotate paper section-by-section. Identify: inputs, outputs, key equations, hyperparameters.	Annotated PDF + algorithm flowchart
4	Environment setup	Clone official repo (if exists). Install dependencies. Run original code ONCE to verify it works.	Working environment + run log

 **Red flag at Week 4:** If original code doesn't run, switch papers immediately.

Month 2: Reproducibility of Selected Paper

Week	Goal	Specific Tasks	Deliverable
5	Re-run & log	Run original code 3x. Record: metrics, runtime, random seeds, hardware specs.	Reproduction log (CSV/table)
6	Sanity checks	Compare your numbers with paper's claims. Are they within $\pm 5\%$? If not, debug.	Comparison table (your's vs. paper)
7	Ablation study (minimal)	Remove ONE component. Does performance drop as paper claims? Proves you understand the mechanism. (if applicable)	Ablation results
8	Reproducibility report	Write 2-page report: What matched? What didn't? Why? (e.g., different GPU, missing details)	Reproducibility Report v1.0

What "reproduction complete" means: Your numbers are within reasonable range (typically $\pm 5\text{-}10\%$ for ML, $\pm 1\text{-}2\%$ for numerical simulations) OR you've documented exactly why they differ.

Month 3: Enhancement Planning & Execution

Week	Goal	Specific Tasks	Deliverable
9	Identify enhancement	Choose ONE from: speed improvement (2x faster), accuracy (+X%), new feature, different dataset, or simpler method. Write hypothesis.	Enhancement proposal (1 page)
10	Baseline for enhancement	Run original code again (fresh) to get baseline for YOUR hardware.	Baseline metrics (fresh)
11	Implement enhancement	Code changes. Run experiments. Keep failed attempts logged.	Working enhanced code
12	Compare & iterate	Run original vs. enhanced side-by-side. Statistical test (t-test or Wilcoxon). Is improvement significant?	Comparative results table

Enhancement tiers (from easiest to hardest):

- ✓ **Tier 1 (safe):** Add visualization, user-friendly CLI, or documentation
 - ✓ **Tier 2 (standard):** Hyperparameter tuning, data augmentation, feature selection
 - ✓ **Tier 3 (ambitious):** Replace one module with a simpler/faster alternative
 - ✓ **Tier 4 (risky):** Novel combination or new dataset (only if original runs early)
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Month 4: Finalization & Documentation

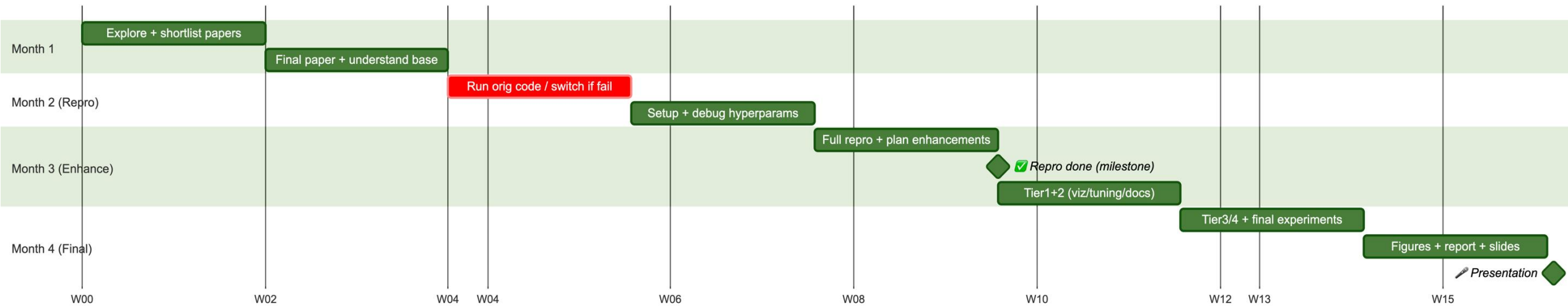
Week	Goal	Specific Tasks	Deliverable
13	Final experiments	Run 5-10 trials for statistical significance. Generate figures (bar charts, learning curves).	Final figures + stats
14	Write final report	Structure: Intro → Reproduction Methodology → Reproducibility Results → Enhancement → Discussion of Differences → Conclusion	Draft report
15	Prepare code repository	Clean code, add README with: "How to reproduce", "How to run enhancement", requirements.txt, license	GitHub repo (public)
16	Final submission + presentation	10-slide deck: (1) Paper summary, (2) Reproduction match/mismatch, (3) Enhancement, (4) Lessons learned	Final submission

Critical Milestones for Assessment (Grade Breakdown Suggestion)

Component	Weight	Due
Paper selection + feasibility	5%	Week 2
Environment runs without error	5%	Week 4
Reproduction log (3 runs)	20%	Week 5
Reproduction report	25%	Week 8
Enhancement proposal	10%	Week 9
Enhancement results (improvement OR learning)	20%	Week 12
Final code + README	10%	Week 15
Final report + presentation	5%	Week 16

PRESENTATION TIMELINE

Bi-weekly Student Project (No Overlap)



Weekly Status Report (for students to submit)

Project: [Paper Title]

Week: [X of 16]

Status: pending / progressing / on track

What I planned to do:

- []

What I actually did:

- []

Reproduction metrics (if applicable):

- Paper's reported: _____

- My reproducibility: _____

- Difference: _____%

Blockers (if any):

- []

Next week's plan:

- []

Supervisor Signature
